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Let $G \neq 1$ be a finite group and let V be an irred rep. Then $\dim V < |G|$.

Proof: Enumerate $G = \{g_i\}_{i \in [n]}$ for $n > 1$ and let $g_1 = 1$. Let $\rho : G \rightarrow GL(V)$ be the homomorphism. If ρ is trivial, then every subspace of V is G -invariant. Then pick $v \in V$ such that $v \neq 0$, so $W = \langle v \rangle$ is a subrep with $\dim W = 1 < n$.

Else, $\rho \neq 1$, so there exist $g_j \in G, v \in V$ such that $g_j \cdot v \neq v$. Consider $w = \sum g_i \cdot v$, which satisfies $h \cdot w = w$ for all $h \in G$. If $w \neq 0$, then $W = \langle w \rangle$ is a subrep. Else, $w = 0$. Consider $W' = \{\sum a_i g_i v : \sum a_i = 0\}$. Note that

$$\sum_{i=1}^n a_i g_i v = \sum_{i=1}^{n-1} a_i g_i v + \left(- \sum_{i=1}^{n-1} a_i \right) \left(- \sum_{i=1}^{n-1} g_i v \right) \in \langle g_i \rangle_{i \in [n-1]},$$

so $\dim W' \leq n - 1$. Further, $g_1 v - g_j v = v - g_j v \neq 0$ is contained in W' , so $W' \neq 0$, so W' is a proper subrep of V .

Let R be a ring and M be a simple R -module. Show that $\text{Hom}_R(M, M)$ is a division ring.

$0 \in \text{Hom}_R(M, M)$, $\text{Hom}_R(M, M)$ is closed under composition, $1_M \in \text{Hom}_R(M, M)$ satisfies $1_M f = f 1_M$ for all $f \in \text{Hom}_R(M, M)$, and function composition is associative.

$(\text{Hom}_R(M, M), +)$ is an abelian group. Left, right distributivity hold by composition rules. Suppose $f \in \text{Hom}_R(M, M)$ is nonzero. Then f is an isomorphism. Pick $g \in \text{Hom}_R(M, M)$ such that $fg = gf = 1_M$, so $\text{Hom}_R(M, M)$ is a division ring.

Sps $\rho : G \rightarrow GL(V)$ is a rep. Suppose $z \in Z(G)$. Then $\rho(z)\rho(g) = \rho(zg) = \rho(gz) = \rho(g)\rho(z)$, so $\rho(z) \in Z(\rho)$.

Sps ρ is irred and $H \leq GL(V)$ is finite abelian.

Let R be a division ring and $H \leq R^\times$ be finite abelian. Then I claim H is cyclic.

Let $K = \mathbb{Z}H$, so $(K, +)$ is abelian, K is closed under $\cdot$, K is abelian, so K is a field.

Let F be a finite field, V/F be finite dimensional, G be finite. Suppose $H \leq \text{Hom}_{FG}(V, V)^\times$ is abelian. Then I claim H is cyclic.

Proof: Since V/F is finite dimensional and F is finite, so is V . A linear map $T : V \rightarrow V$ is determined by the image of the elements of a basis of V (which has finitely many elements), so $\text{Hom}_{FG}(V, V)$ is finite.

Let $K = \mathbb{Z}H/(\mathbb{Z}H)^\times$. Since H is commutative, so is $\mathbb{Z}H$. Since $\text{Hom}_{FG}(V, V)$ is a division ring, $\text{Hom}_{FG}(V, V)$ has no zero divisors, so $\mathbb{Z}H$ is an integral domain. Hence, $K \leq \text{Hom}_{FG}(V, V)$ is a finite field.

It follows that $K^\times$ is cyclic. Since $H \leq K^\times$, H is also cyclic.

Suppose that $\varphi : G \rightarrow GL(V)$ is faithful irreducible. Then $Z(G)$ is cyclic.

Proof: $\varphi(Z(G)) \cong Z(G)$ (since φ is faithful) is a finite abelian subgroup of $\text{Hom}_{FG}(V, V)^\times$. By the previous result, $\varphi(Z(G))$ is cyclic, so $Z(G)$ is also cyclic.

Suppose finite abelian noncyclic G acts on V , an elementary abelian p -group (over $\mathbb{F}_p$). If $W \leq V$ is an irred, then there exists $g \in G$ such that $W \leq C_V(g)$.

Proof: Suppose $W \cap C_V(g) = 0$ for all nontrivial $g \in G$. Then $g \cdot w \neq w$ for all $w \neq 0$ in W , so in particular, $\varphi : G \rightarrow W$ is faithful. It follows $Z(G) = G$ is cyclic, absurd.

Hence, $W \cap C_V(g) \neq 0$ for some nontrivial $g \in G$. Since W is an irred, we have $W \leq C_V(g)$.

Assume further that $\gcd(p, |G|) = 1$. Then V decomposes as the sum of irreps $V = \bigoplus_{i \in [n]} W_i$. For each $i \in [n]$, pick nontrivial $g_i \in G$ such that $W_i \leq C_V(g_i)$, so $V = \bigcup_{i \in [n]} C_V(g_i)$, as desired.